MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 8613

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen, ceramic-metal

See figure 1

Mounting position: Any

Weight: 1 pound 2 ounces (510.3 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	egy	egx	Ecc	dik/dt	Ib
Unit:	Vac	kv	kv	kVdc	v	v	Vdc	a/us	a
Maximum:	6.8 (Note 4)	16.0 (Note 1)	16.0 (Note 2)	---	600 (Note 3)	200	200	2,000	500
Minimum:	5.8	2.0	5% epy	1.0	175	---	---	---	---
TEST CONDITIONS:	6.3	---	---	---	175	---	0	---	---

ABSOLUTE RATINGS:

Parameter:	Ip	Ib	tk	prr	Pb	tj	TA	Cooling
Unit:	Aac	Adc	sec	---	---	us	°C	---
Maximum:	8.0	0.5	---	---	10.0×10^9	0.005	125	Note 5
Minimum:	---	---	180	---	---	Note 6	---	---
TEST CONDITIONS:	---	---	180	2,000	---	---	Ambient	---

GENERAL:

Qualification - Required

① denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3267	Instantaneous starting	9,10	epy = 16.0 kv (min); Ef = 6.8 Vac	0.65	II	---	---	---	---
(B) 3246	Operation (1)	4,9, 11	epy = 18.0 kv; Ef = 5.8 Vac	0.65	II	egy	---	175	v
3256	Anode delay time	-	Operation (1); t = 120	0.65	II	tad	---	0.50	µs
3256	Anode delay time drift	12	Anode delay time	0.65	II	Δtad	---	0.10	µs
3241	Heater current	-	Ef = 6.3 Vac	0.65	II	If	6.0	11.5	Aac
(B) 3247	DC anode voltage for conduction	-	Ef = 5.8 Vac	0.65	II	Ebb	---	1,000	Vdc
3251	Pulse emission	-	ik = 500 a; tp = 5.0 µs ±10%; tr = 0.5 µs (max); prr = 60 ±10%; time interval = 2.5 µs	0.65	II	egk	---	175	v
	Quality conformance inspection, part 2								
1031	Sweep-frequency vibration	8	10 to 2,000 Hz	---	---	---	---	---	---
---	Sweep-frequency vibration end points:								
3246	Operation (1)	-		---	---	egy	---	175	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	1,000	Vdc
3261	Time jitter	-		---	---	tj	---	0.005	µs
3246	Operation (2)	11	Operation (1); Ef = 6.8 Vac; tk = 180 seconds	---	---	egy	---	175	v
(B) 3246	Operation (3)	14	epy = 14.0 kv; prr = 2,500; Ef = 5.8 Vac; t = 300 seconds	---	---	egy	---	175	v
3261	Time jitter	13	Operation (1), except epy = 8 kv	---	---	tj	---	0.005	µs
	Quality conformance inspection, part 3								
---	Life test	9	Group C; operation (1); t = 1,000 hours	---	---	---	---	---	---
---	Life-test end points:								
3246	Operation (1) and (2)	-		---	---	egy	---	200	v
3256	Anode delay time	-		---	---	tad	---	0.6	µs
3247	DC anode voltage for conduction	-	egy = 200 v	---	---	Ebb	---	1,250	Vdc
3261	Time jitter	-	egy = 200 v	---	---	tj	---	0.010	µs
1041	Shock	7	100 G	---	---	---	---	---	---
---	Shock test end points:								
3246	Operation (1)	-		---	---	egy	---	175	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	1,000	Vdc
3261	Time jitter	-		---	---	tj	---	0.005	µs
(B) 3246	Operation at elevated ambient temperature	7,9, 15	TA = 125°C; t = 5 hours	---	---	egy	---	175	v

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NOTES:

1. Instantaneous starting is permissible. The maximum permissible instantaneously applied epy is 16.0 kv and shall not be attained in less than 0.04 second.

2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 µs (maximum) duration, shall not exceed 5.0 kv during the first 25 µs following the anode pulse.

3. The driver pulse, measured at tube socket with thyratron grid disconnected, shall have the following characteristics: amplitude per ratings; tr = 0.35 µs (maximum); tp = 2.0 µs (minimum); Zg = 250 to 500 ohms. At -55°C, 200 V (minimum) shall be required.

4. The optimum reservior voltage for operation in accordance with operation (1) conditions shall be 6.3 Vac and shall be held to within ±7-1/2 percent.

5. A cooling airblast of 10 cubic feet per minute (cfm) may be directed into the anode cup when operating at maximum anode dissipation.

6. Appreciably less jitter than 0.005 µs can be realized if the anode voltage is 8.0 kv or more, the grid-drive amplitude is near the maximum and the grid-drive impedance is near minimum.

(B) 7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

8. There shall be no pronounced resonance in the range from 10 to 2,000 Hz.

9. The circuit constants shall be chosen under resonant charging conditions so that : epy = 18.0 kv; ib = 180 a (minimum); $\frac{dik}{dk}$ = 1,500 a/µs (minimum); tp = 1.0 ±10 percent µs; prr = 1,000 (minimum).

   Grid pulse characteristics shall be:tr = 0.35 µs (minimum); tp = 2.0 µs (maximum); and driver impedance = 500 ohms (minimum).

10. The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 16.0 kv (minimum) within 0.03 second. (The filter in the rectifier shall be designed so that the epy reaches at least 7.0 kv within 0.015 second).

11. The tube shall operate continuously for 10 minutes.

12. This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 and 10 minutes of the operation (1) test. The change in anode delay time (with respect to the 2-minute reading) shall not exceed the value specified herein at any time during this test.

13. The tube shall be tested by applying a peak forward anode voltage not to exceed that specified in the test conditions for the time jitter test immediately after the cathode warmup period (tk). The variation in firing time (tj), shall be not greater than the amount specified herein after 60 seconds of operation.

14. The circuit constants shall be so chosen that the epy = 14.0 kv; ib = 130 a (minimum); $\frac{dik}{dt}$ = 1,250 a/µs (minimum); tp = 0.4 µs ±10 percent; prr = 2,500 (minimum). Grid pulse shall be the same as note 9.

15. This test shall be conducted for a total of 5 consecutive hours with no more than three kickouts and with no evidence of detrimental anode heating. The tube shall be started with Ef = 107-1/2 percent Vac and operate at this value for 4 hours. At the start of the fifth hour and while the tube is still operating, the filament voltage shall be lowered to Ef = 92-1/2 percent Vac and remain there for the final hour of operation.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

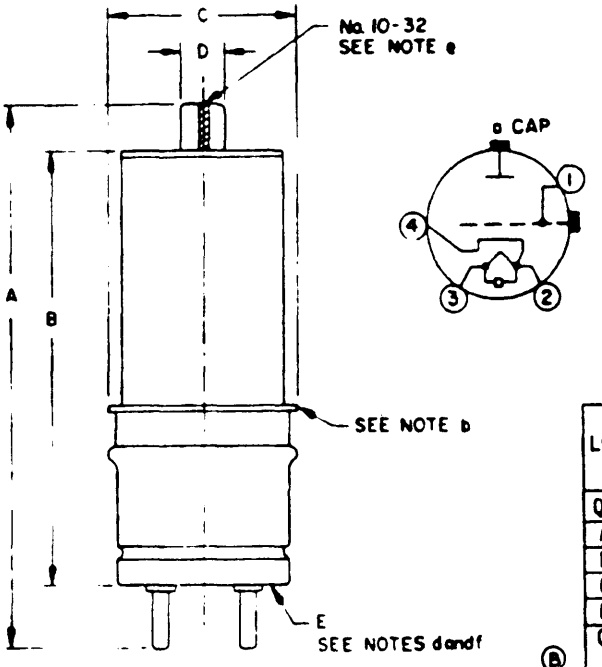
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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PIN CONNECTIONS	
Pin No.	Element
1	g
2	h, k, r
3	k, r
4	k
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.125 (155.58)	6.625 (168.28)
B	4.750 (120.65)	5.150 (130.81)
C	2.000 (50.80)	2.375 (60.33)
D	.559 (14.20)	.573 (14.55)
Quality conformance inspection, part 3 (SEE NOTE f)		
E	Base: A4-18 (EIA)	

NOTES:
a. Do not use metal clamp on ceramic envelope.
b. This flange is mechanically and electrically connected to the grid.
c. Recommended anode connector - lightweight spring-clip type (National Co. type 12, or equivalent).
d. The cathode shall be isolated from the base.
e. In certain applications it may be necessary to extend the anode connector as follows:
   To replace a 6587 tube type with an 8613, screw the anode adapter marked 6587 firmly into the 8613 anode connector (see figure 1A).
   To replace a 5C22 tube type with an 8613, screw the anode adapter marked 5C22 firmly into the 8613 anode connector (see figure 1B).
f. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8613.

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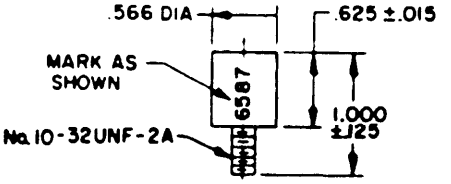


FIGURE 1A

INCHES	MM
.015	.38
.125	3.18
.566	14.38
.625	15.88
1.000	25.40
2.125	53.98
2.500	63.50

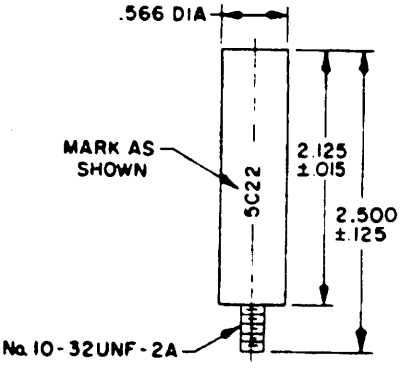


FIGURE 1B

NOTE:
Unless otherwise specified, tolerance is ±.005 (.13 mm).

FIGURE 1. Outline drawing of electron tube type 8613 - Continued.

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reference	specified		Resistor	Rated

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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